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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 39.4

$$\begin{aligned}
 & \int \frac{3x+1}{(x^2-4x+5)^2} dx \\
 &= \int \frac{3x+1}{((x-2)^2+1)^2} \\
 & t = x - 2 \Rightarrow dt = dx \\
 & \Rightarrow \int \frac{3t+7}{(t^2+1)^2} dt \\
 &= 3 \int \frac{t}{(t^2+1)^2} dt + 7 \int \frac{dt}{(t^2+1)^2} \\
 & A = 3 \int \frac{t}{(t^2+1)^2} dt = \frac{3}{2} \int (t^2+1)^{-2} d(t^2+1) = -\frac{3}{2(t^2+1)} \\
 & B = 7 \int \frac{1}{(t^2+1)^2} dt = 7 \left(\frac{t}{2(t^2+1)} + \frac{1}{2} \int \frac{1}{t^2+1} dt \right) \\
 &= \frac{7t}{2(t^2+1)} + \frac{7}{2} \text{Bgtan}(t) \\
 & A + B = \frac{7t-3}{2(t^2+1)} + \frac{7}{2} \text{Bgtan}(t) + C \\
 &= \frac{7(x-2)-3}{2((x-2)^2+1)} + \frac{7}{2} \text{Bgtan}(x-2) + C \\
 &= \frac{7x-17}{2(x^2-4x+5)} + \frac{7}{2} \text{Bgtan}(x-2) + C
 \end{aligned}$$

References